BreakCheck

If I ship this API version, who breaks?



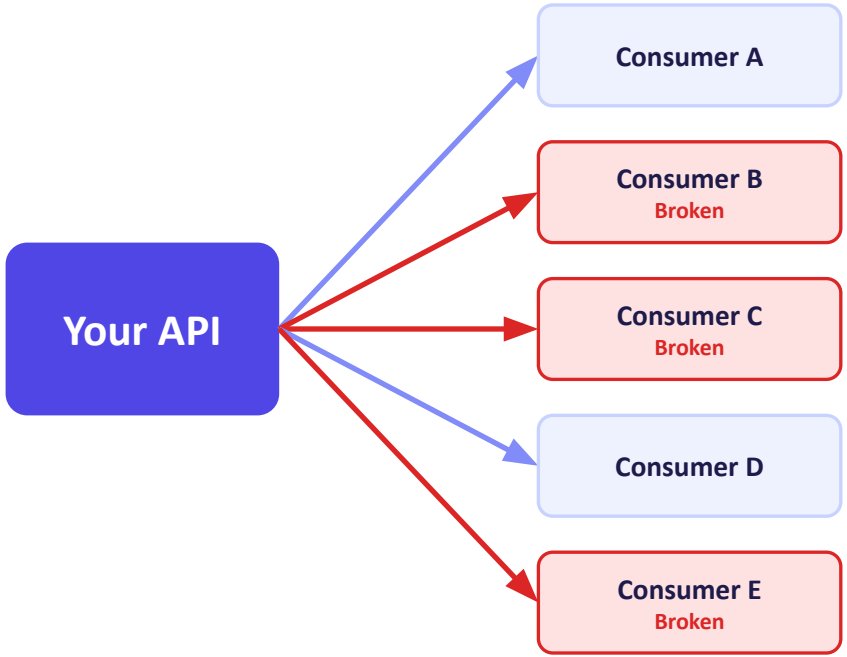
Team Almost Avengers, SVNIT

Harsheel Kumar Sahoo (Lead) · Dhruv Papaiya · Tanishka Jethani

NEXUS UNCHARTED · MongoDB Student Hackathon · Track 3: Developer Tools

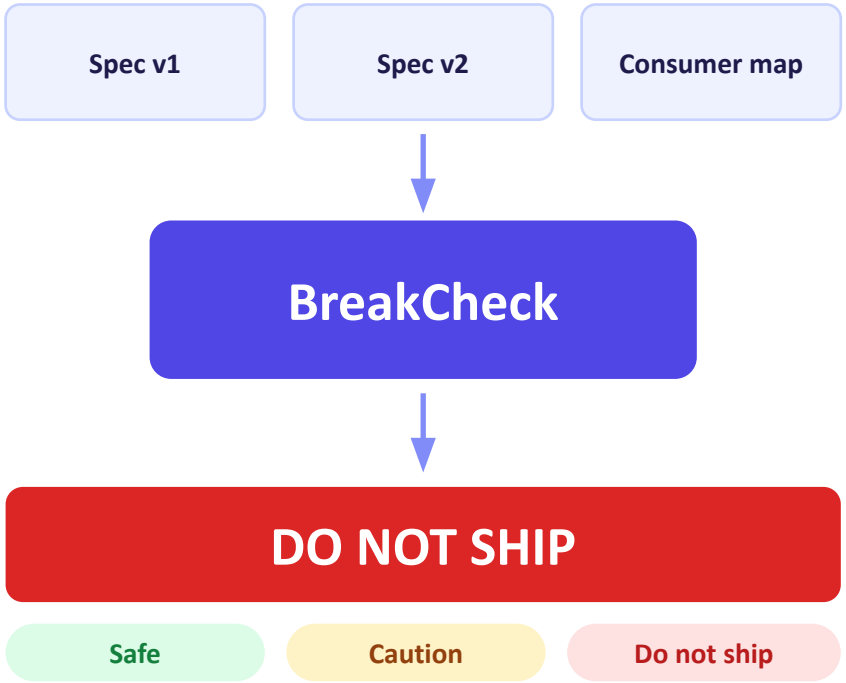
The Problem

- APIs change constantly
- Nobody knows who depends on what
- Breakage is found after release
- Diff tools show text, not impact



Our Solution

- Compare two OpenAPI versions
- Match changes to what each consumer actually uses
- Verdict: Safe / Caution / Do not ship
- Migration task per team



How It Works

1

Upload

Two specs plus a consumer map

2

Flatten

Each spec becomes keyed elements, ready to compare

3

Classify

One rule table sorts every change

- **Breaking**
- **Potential**
- **Safe**

4

Join

MongoDB **$lookup** joins changes to consumers

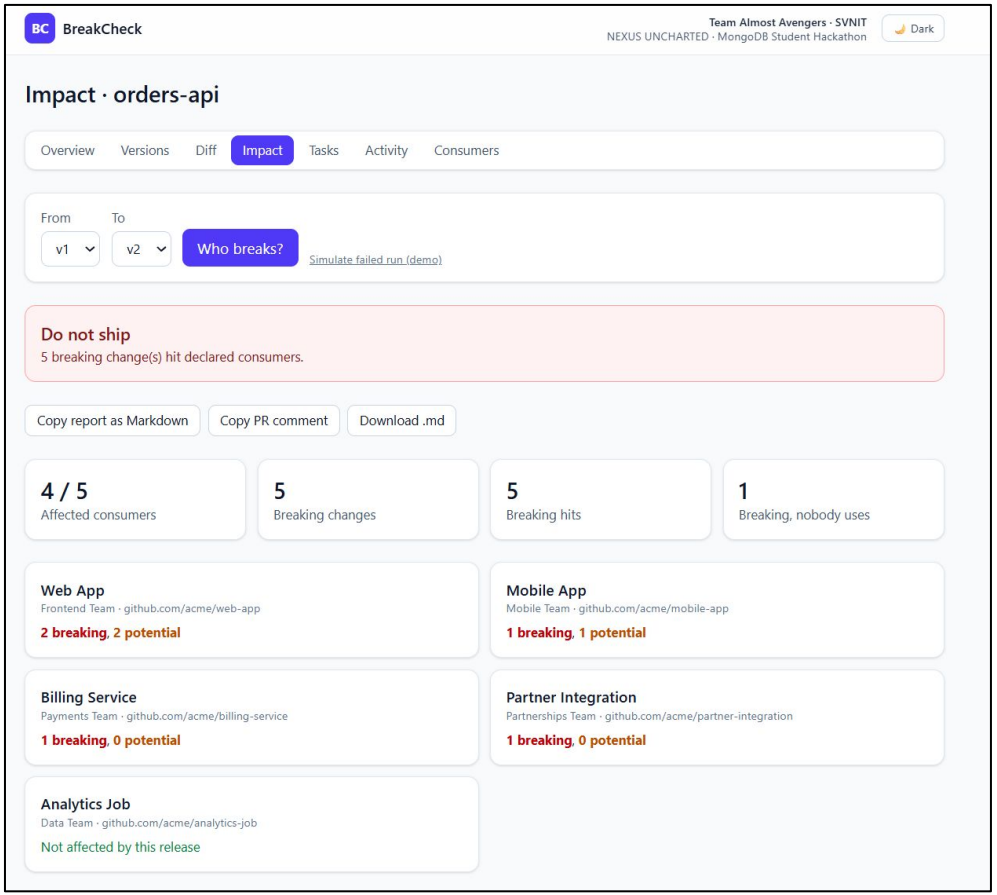
Result: v1 → v2

5	3	11
Breaking	Potential	Safe

Verdict: DO NOT SHIP

- Web App, Billing, Mobile, Partner break
- Analytics Job not flagged

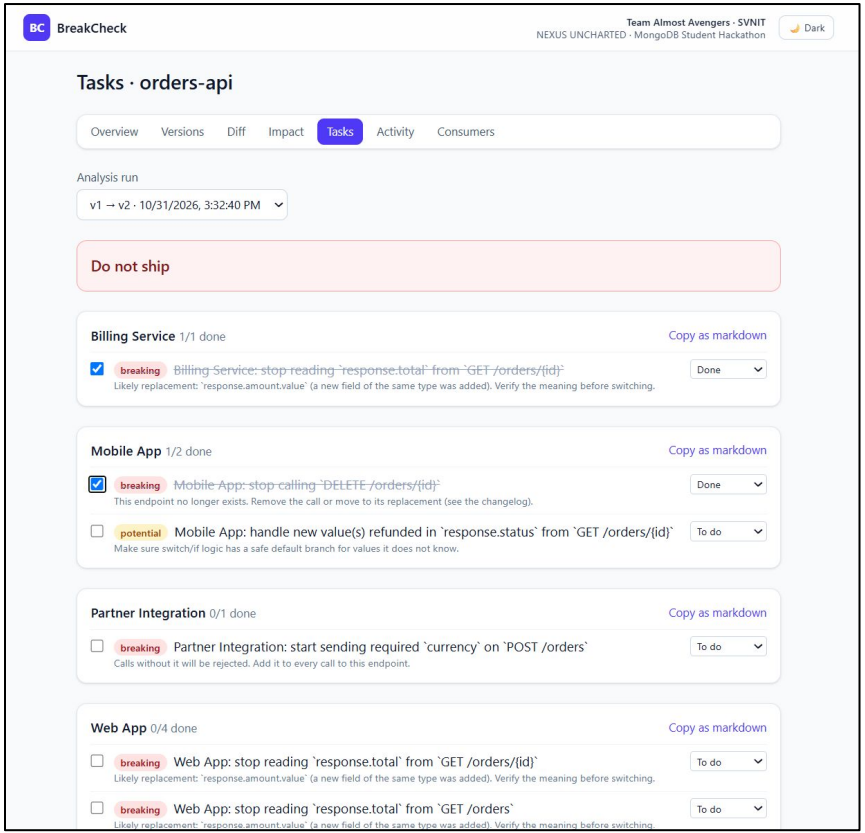




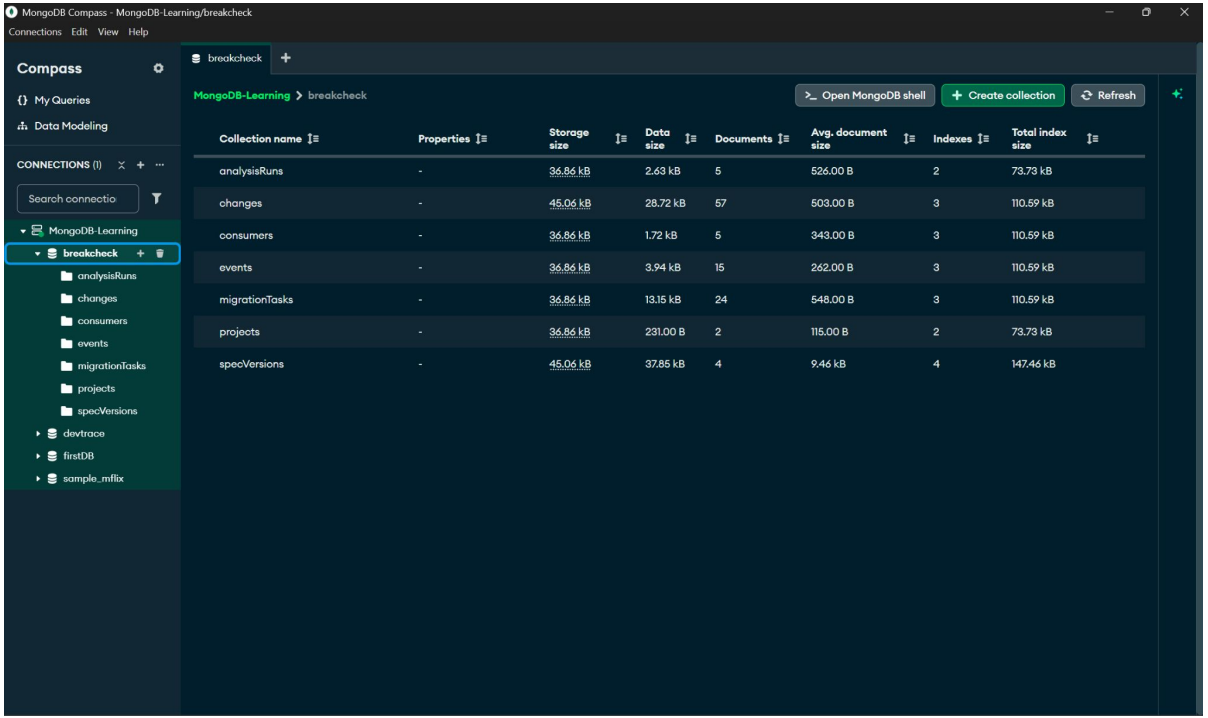
Migration Tasks

8 **tasks generated for v1 → v2**

- Fix hint per change
- Status saved in MongoDB
- Survives re-runs



MongoDB Data Model



7 Collections

- analysisRuns
- changes
- consumers
- events
- migrationTasks
- projects
- specVersions

- $jsonSchema on required fields
- Raw specs and payloads stay flexible

Indexes and Aggregations

8 / 8 queries use IXSCAN in **explain()**

- Each index serves one screen

Aggregations

- **Impact join**
- **Summary**
- **Risk ranking**
- **Trend**

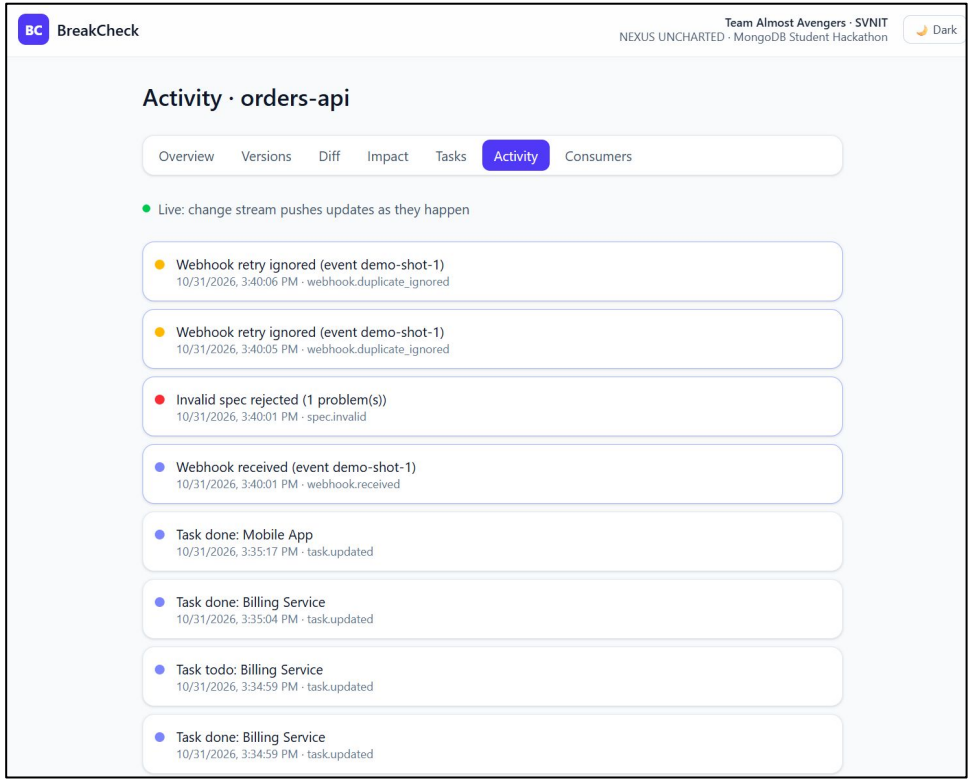
```
C:\Users\Harsheel Kumar Sahoo\Desktop\breakcheck>npm run explain

> breakcheck@0.1.0 explain
> tsx scripts/explain-indexes.ts

◇ injected env (2) from .env.local
PASS Diff page: breaking changes of a version
     index: changes (toVersionId, severity)
     plan: IXSCAN  keysExamined=5  docsExamined=5  returned=5
PASS Endpoint history
     index: changes (projectId, method, path)
     plan: IXSCAN  keysExamined=5  docsExamined=5  returned=5
PASS Latest run for a project
     index: analysisRuns (projectId, startedAt desc)
     plan: IXSCAN  keysExamined=1  docsExamined=1  returned=1
PASS Open tasks per consumer
     index: migrationTasks (consumerId, status)
     plan: IXSCAN  keysExamined=0  docsExamined=0  returned=0
PASS All tasks of a run
     index: migrationTasks (runId)
     plan: IXSCAN  keysExamined=8  docsExamined=8  returned=8
PASS Activity feed
     index: events (projectId, receivedAt desc)
     plan: IXSCAN  keysExamined=9  docsExamined=9  returned=9
PASS Duplicate spec check
     index: specVersions unique (projectId, checksum)
     plan: IXSCAN  keysExamined=0  docsExamined=0  returned=0
PASS Webhook duplicate check
     index: events unique (eventId)
     plan: IXSCAN  keysExamined=0  docsExamined=0  returned=0

All queries use an index
```

Idempotent by Design



Unique indexes

On checksum and eventId

Stable hash _id

For changes and tasks

Re-diff creates 0 duplicates

Same input, same documents

Live Activity Feed

One change stream

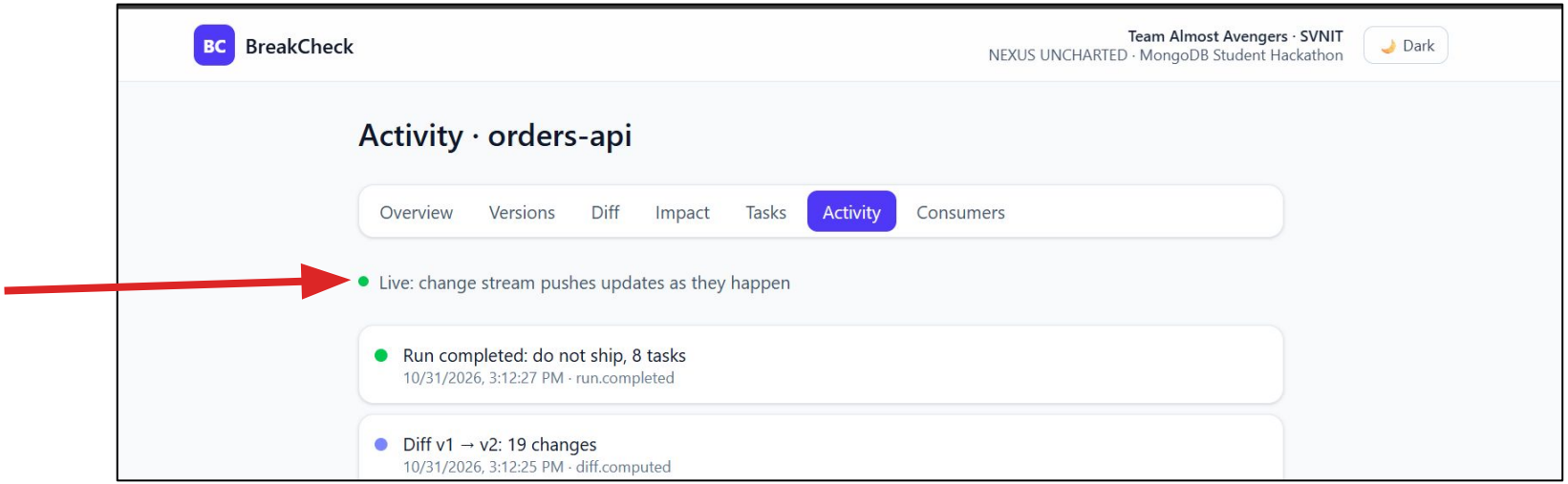
on **events**

Pushed over SSE

to browsers

Polling fallback

on serverless hosts



Built for CI

- CLI needs no database
- Shares code with the web app
- 53 unit tests

Exit codes

0	Safe
1	Breaks
2	Bad input

```
C:\Users\Harsheel Kumar Sahoo\Desktop\breakcheck>npx tsx cli/breakcheck.ts compare samples/orders-v1.yaml samples/orders-v2.yaml --consumers sampl
es/consumers.json

BreakCheck: samples/orders-v1.yaml -> samples/orders-v2.yaml
19 changes: 5 breaking, 3 potential, 11 safe

Mobile App (1 breaking, 1 potential)
  [breaking] Mobile App: stop calling `DELETE /orders/{id}`
      This endpoint no longer exists. Remove the call or move to its replacement (see the changelog).
  [potential] Mobile App: handle new value(s) refunded in `response.status` from `GET /orders/{id}`
      Make sure switch/if logic has a safe default branch for values it does not know.

Web App (2 breaking, 2 potential)
  [potential] Web App: handle new value(s) refunded in `response.status` from `GET /orders`
      Make sure switch/if logic has a safe default branch for values it does not know.
  [breaking] Web App: stop reading `response.total` from `GET /orders`
      Likely replacement: `response.amount.value` (a new field of the same type was added). Verify the meaning before switching.
  [potential] Web App: handle new value(s) refunded in `response.status` from `GET /orders/{id}`
      Make sure switch/if logic has a safe default branch for values it does not know.
  [breaking] Web App: stop reading `response.total` from `GET /orders/{id}`
      Likely replacement: `response.amount.value` (a new field of the same type was added). Verify the meaning before switching.

Billing Service (1 breaking, 0 potential)
  [breaking] Billing Service: stop reading `response.total` from `GET /orders/{id}`
      Likely replacement: `response.amount.value` (a new field of the same type was added). Verify the meaning before switching.

Partner Integration (1 breaking, 0 potential)
  [breaking] Partner Integration: start sending required `currency` on `POST /orders`
      Calls without it will be rejected. Add it to every call to this endpoint.

=> DO NOT SHIP: 5 breaking change(s) hit declared consumers.


C:\Users\Harsheel Kumar Sahoo\Desktop\breakcheck>   echo %ERRORLEVEL%
1
```

Failures Handled

Invalid spec

Exact location of the error

Duplicates

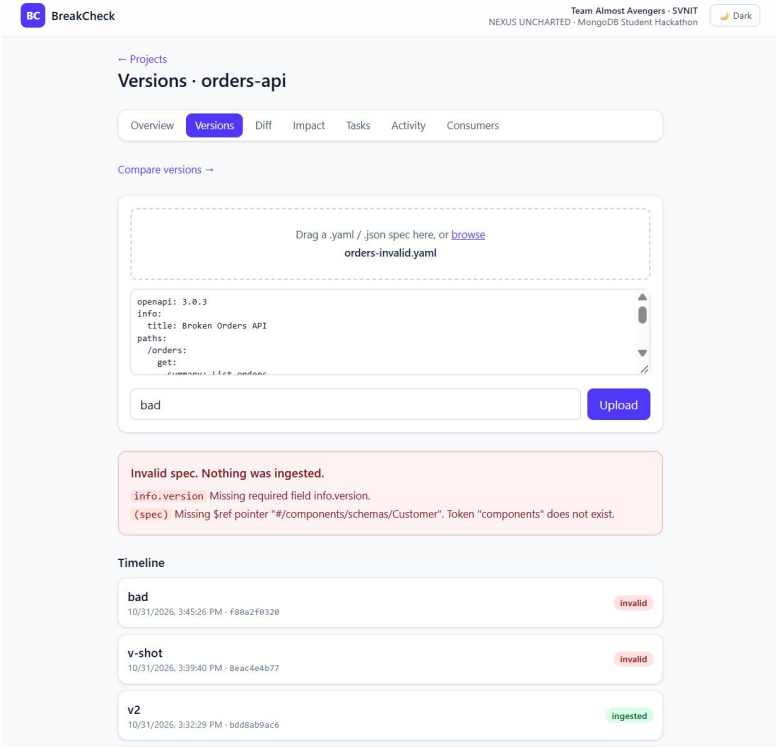
Duplicate upload and webhook ignored

Incomplete consumer map

Warnings raised

Failed job

Stored with its error



Limits and Next

Limits

- Consumer map is declared, not discovered
- oneOf / anyOf not expanded yet

Next

- GitHub Action
- Gateway-log discovery

Thank You

Team Almost Avengers · SVNIT

Live Link

breakcheck-live.vercel.app

GitHub

github.com/Harsheel-Kumar-Sahoo/BreakCheck